ELEC3104: Digital Signal Processing

Topic 5: Digital filter Design

Model Answers to Tutorial Questions

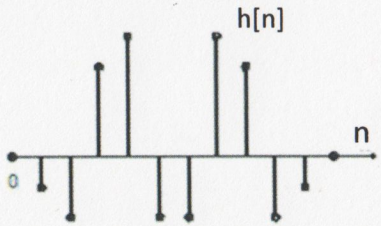
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Topic 5: Tutorial Questions (SOLO -Level 1)

SOLO – Level 1

1. The impulse response h[n] of a digital filter is shown below. What can be said about this filter?



I) It is an FIR filter; ✓
II) It is an IIR filter;
III) It has linear phase; ✓
IV) It is a low-pass filter.

Tick the correct answer:

a) I
✓b) I, III
c) II, III
d) I, III, IV
e) None of the above

- The coefficients are symmetrical ⇒ linear phase
- Also finitie number of coefficients

2. One of the major problem that you may encounter when designing digital IIR filters using impulse invariant transformation is: (Tick the correct answer)

a) Mapping the analogue impulse response h(t) to its sampled version h[n];
✓b) Effects of Aliasing ; Increase the sampling rate to avoid aliasing
c) Inability to map the poles of the analogue fitters to the corresponding digital filters;
d) None of the above.

Topic 5: Tutorial Questions (SOLO -Level 1)

SOLO – Level 1

3. Window functions are used to truncate a signal to produce a signal of finite duration. The truncation process: (Tick the correct answers)

✓a) Causes frequency distortion in the output spectrum;
✓b) Creates ripples in the output spectrum;
✓c) Convolves the input spectrum with the window function spectrum;
d) None of the above.

} see lecture notes Page 28 (Topic 5)

4. The bilinear transform is used for: (Tick the correct answers)

✓a) Mapping the left half of s plane into the unit circle in the z-plane thus avoiding aliasing of frequency components;
✓b) Transforming a stable analogue filter into a stable digital filter;
c) Avoiding non-linear mapping;
d) None of the above.

} see lecture notes Page 13 (Topic 5)

5. Which digital filter design method avoids aliasing but requires 'pre-warping' of the filter specifications ?

a) Frequency sampling;
b) Windowing;
✓c) Bi-linear transform;
d) Impulse invariant method.

} (See Lecture notes Page 16 (Topic 5)

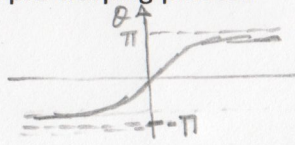
Topic 5: Tutorial Questions (SOLO-Level 2)

SOLO – Level 2

1. (a) When using the bilinear transformation to design a recursive digital filter from an analogue filter, pre-warping of the analogue filter's frequency response is required. State the reason for this pre-warping process.

$\omega_c = \frac{2}{T}\tan\left(\frac{\theta}{2}\right)$

↑ analogue frequency



This mapping is highly non-linear therefore pre-warping of the analogue filter's response is required

(b) Use the bilinear z-transform to design an IIR digital lowpass filter based on the prototype function G(s): $G(s) = \frac{1}{s^2+s+1}$; The digital filter is to be designed to operate at a sampling rate of 8 kHz with a 3dB cut-off frequency at 2 kHz. Show that G(z) reduces to $G(z) = \frac{(z+1)^2}{(3z^2+1)}$.

We must prewarp the cut-off frequency ω_c

$\omega_c' = \frac{2}{T}\tan\left(\frac{\theta}{2}\right) = \frac{2}{T}\tan\left(\frac{1}{2}\left(2\pi\frac{2}{8}\right)\right) = \frac{2}{T}$ ← θ

$G(s) = \frac{1}{s^2+s+1}$; The denormalised analogue transfer function is:

$G_1(s) = G(s)\Big|_{s=\frac{s}{\omega_c'}} = \frac{1}{\left(\frac{s}{\omega_c'}\right)^2+\left(\frac{s}{\omega_c'}\right)+1} = \frac{(\omega_c')^2}{s^2+\omega_c's+(\omega_c')^2}$

$\therefore G(z) = G_1(s)\Big|_{z=\frac{2}{T}\left(\frac{z-1}{z+1}\right)} = \frac{\left(\frac{2}{T}\right)^2}{\left[\frac{2}{T}\frac{z-1}{z+1}\right]^2+\frac{2}{T}\left(\frac{2}{T}\frac{z-1}{z+1}\right)+\left(\frac{2}{T}\right)^2}$

$\therefore G(z) = \frac{1}{\left(\frac{z-1}{z+1}\right)^2+\left(\frac{z-1}{z+1}\right)+1} = \frac{(z+1)^2}{(z^2-2z+1)+(z^2-1)+z^2+2z+1}$

$= 3z^2+1$

$\therefore G(z) = \frac{(z+1)^2}{3z^2+1}$

2. Use the bilinear z-transform to design an IIR digital lowpass filter based on the prototype function $G(s) = \frac{2}{s+2}$. The digital filter is to be designed to operate at a sampling rate of 8 kHz with a 3 dB cut-off frequency at 2kHz. Determine the transfer function G(z).

We must Prewarp the cut-off frequency ω_c.

$\omega_c' = \frac{2}{T}\tan\left(\frac{\theta}{2}\right) = \frac{2}{T}\tan\left(\frac{1}{2}2\pi\frac{f_a}{f_s}\right) = \frac{2}{T}\tan\left(\frac{1}{2}\times 2\pi\times\frac{2}{8}\right) = \frac{2}{T}$

$G(s) = \frac{2}{s+2}$ let $\omega_c = 2$ $\therefore G_1(s) = \frac{\omega_c'}{s+\omega_c'}$

$G(z) = G(s)\Big|_{s=\frac{2}{T}\left(\frac{1-z^{-1}}{1+z^{-1}}\right)} = \frac{\omega_c'}{\frac{2}{T}\left(\frac{1-z^{-1}}{1+z^{-1}}\right)+\omega_c'} = \frac{T\omega_c'(1+z^{-1})}{2-2z^{-1}+T\omega_c'+T\omega_c'z^{-1}}$

$G(z) = \frac{T\cdot\omega_c'}{2+T\omega_c'}\cdot\frac{1+z^{-1}}{1-\frac{2-T\omega_c'}{2+T\omega_c'}z^{-1}}$

$= \frac{T\cdot\frac{2}{T}}{2+T\cdot\frac{2}{T}}\cdot\frac{1+z^{-1}}{1+\left(\frac{2-T\cdot\frac{2}{T}}{2+T\cdot\frac{2}{T}}\right)z^{-1}} = \frac{1}{2}\cdot\frac{1+z^{-1}}{1+0}$

$G_1(z) = \frac{1}{2}\cdot(1+z^{-1})$

SOLO – Level 2

3. Convert $H(s) = \frac{4s+7}{s^2+5s+4}$ to $H(z)$ using impulse invariant transformation, where the sampling frequency is 2Hz.

   [**Hint:** Find the partial fraction for H(s) first, before applying impulse invariant transformation]

$H(s) = \frac{4s+7}{(s+1)(s+4)} = \frac{A}{s+4} + \frac{B}{s+1} = \frac{3}{s+4} + \frac{1}{s+1}$

$T = \frac{1}{2} = 0.5s$

Impulse invariant transformation:

$\frac{1}{s+a} \rightarrow \frac{1}{1-e^{-aT}z^{-1}}$

$H(z) = \frac{3}{1-e^{-4T}z^{-1}} + \frac{1}{1-e^{-T}z^{-1}} = \frac{3}{1-e^{-2}z^{-1}} + \frac{1}{1-e^{-0.5}z^{-1}}$

$H(z) = \frac{3(1-e^{-0.5}z^{-1}) + (1-e^{-2}z^{-1})}{(1-e^{-2}z^{-1})(1-e^{-0.5}z^{-1})} = \frac{4 - 1.9549z^{-1}}{1 - 0.7419z^{-1} + 0.0821z^{-2}}$

4. (a) A second-order analogue band pass filter can be designed from a first-order normalised analogue low pass Butterworth filter $H(s) = \frac{1}{s+1}$ by performing the following low pass to band pass transformation:

$$s \rightarrow \frac{s^2 + \omega_0^2}{B\,s}; \quad B = \omega_u - \omega_l; \quad \omega_0^2 = \omega_u \omega_l$$

where ω_u is the upper cut-off frequency and ω_l is the lower cut-off frequency. Show that the analogue bandpass filter obtained using the above transformation is given by

$$G(s) = \frac{Bs}{s^2 + Bs + \omega_0^2}$$

(b) An IIR digital bandpass filter G(z) is to be designed to approximate the above analogue transfer function G(s). The digital bandpass filter is to operate at a sampling rate of 2 kHz with a lower cut-off frequency (f_l)at 200 Hz and the upper cut-off frequency (f_u) at 300 Hz. Using the bilinear transformation method, obtain the transfer function G(z) of the digital filter and show that G(z) reduces to

$$G(z) = 0.1367\frac{(1 - z^{-2})}{1 - 1.2362z^{-1} + 0.7265z^{-2}}$$

(a) $H(s) = \frac{1}{s+1} \Rightarrow G(s) = H(s)\Big|_{s \rightarrow \frac{s^2+\omega_0^2}{Bs}} = \frac{1}{\frac{s^2+\omega_0^2}{Bs} + 1} = \frac{Bs}{s^2 + Bs + \omega_0^2}$

(b) $f_u = 300$ Hz, $f_L = 200$ Hz

The prewarped passband edge frequencies are

$\omega_L' = \frac{2}{T}\tan\left(\frac{\omega_L T}{2}\right) = \frac{2}{T}\tan\left(2\pi\frac{200}{2000}\times\frac{1}{2}\right) = \frac{2}{T}\times 0.3249$

Similarly $\omega_u' = \frac{2}{T}\tan\left(2\pi\frac{300}{2000}\times\frac{1}{2}\right) = \frac{2}{T}\times 0.5095$

$\therefore \omega_0^2 = \omega_L' \cdot \omega_u' = \left(\frac{2}{T}\right)^2 \times 0.1655$; $B = \omega_u' - \omega_L' = \frac{2}{T}(0.1846)$

(b) $G(z) = G(s)\Big|_{s=\frac{2}{T}\frac{z-1}{z+1}}$

$$= \frac{\frac{2}{T}(0.1846) \times \frac{2}{T}\frac{z-1}{z+1}}{\left(\frac{2}{T}\frac{z-1}{z+1}\right)^2 + \frac{2}{T}0.1846\left(\frac{2}{T}\frac{z-1}{z+1}\right) + \left(\frac{2}{T}\right)^2 \cdot 0.1655}$$

$$= \frac{0.1846\,\frac{z-1}{z+1}}{\left(\frac{z-1}{z+1}\right)^2 + 0.1846\left(\frac{z-1}{z+1}\right) + 0.1655}$$

$$= \frac{0.1846\,(z+1)(z-1)}{(z^2 - 2z + 1) + 0.1846z^2 - 0.1846 + 0.1655(z^2 + 2z + 1)}$$

$G(z) = \frac{0.1367(1 - z^{-2})}{1 - 1.2362z^{-1} + 0.7265z^{-2}}$

Topic 5: Tutorial Questions (SOLO-Level 2)

SOLO – Level 2

5. You are required to design a linear phase FIR low pass filter. The length of the impulse response is set to N = 7 and the coefficients are to be determined using a windowing technique. An ideal low pass filter $H_d(\theta)$, with linear phase can be characterised in the frequency domain by

$$H_d(\theta) = \begin{cases} e^{-j\beta\theta} & 0 \le |\theta| \le \frac{\pi}{2} \\ 0 & \text{otherwise} \end{cases}$$

(i) Determine β .

(ii) Determine the impulse response $h_d[n]$.

(i) $\beta = \frac{N-1}{2} = \frac{7-1}{2} = 3$

For a linear phase filter symmetrical coefficients centred around 3.

(ii) $h_d[n] = \frac{1}{2\pi} \int_{-\pi/2}^{\pi/2} e^{-j3\theta} e^{jn\theta} d\theta$

$= \frac{1}{2\pi} \left[\frac{e^{j(n-3)\theta}}{(n-3)j} \right]_{-\pi/2}^{\pi/2}$

$h_d[n] = \frac{1}{2\pi j(n-3)} \left[e^{j(n-3)\pi/2} - e^{-j(n-3)\pi/2} \right]$

$= \frac{1}{\pi(n-3)} \left(\frac{e^{j(n-3)\pi/2} - e^{-j(n-3)\pi/2}}{2j} \right)$

$= \sin(n-3)\frac{\pi}{2}$

$\therefore h_d[n] = \frac{\sin(n-3)\frac{\pi}{2}}{\pi(n-3)}$

$h_d[3] = \frac{1}{2}$ [for $n=3$]

Topic 5: Tutorial questions (SOLO - Level 3)

SOLO – Level 3

1. (a) A second-order analogue band pass filter with an s-domain transfer function is given by

$$H(s) = \frac{b_p s}{s^2 + b_p s + \omega_p^2} \qquad (1)$$

Where ω_p and b_p are the centre frequency and bandwidth of the filter respectively, both expressed in rad/s. By applying the bilinear transformation to equation (1) a digital filter with the following transfer function can be obtained:

$$H(z) = \frac{a_0 - a_1 z^{-2}}{1 + b_1 z^{-1} + b_2 z^{-2}} \qquad (2)$$

Show that the digital filter coefficients are given by

$$a_0 = a_1 = \frac{2b_pT}{4 + 2b_pT + \omega_p^2 T^2}; \quad b_1 = \frac{2\omega_p^2 T^2 - 8}{4 + 2b_pT + \omega_p^2 T^2}; \quad b_2 = \frac{4 + \omega_p^2 T^2 - 2b_pT}{4 + 2b_pT + \omega_p^2 T^2}$$

$$H(s) = \frac{b_p s}{s^2 + b_p s + \omega_p^2}$$

$$\therefore H(z) = \frac{b_p \cdot \frac{2}{T}\left(\frac{1 - z^{-1}}{1 + z^{-1}}\right)}{\left(\frac{2}{T}\right)^2\left(\frac{1 - z^{-1}}{1 + z^{-1}}\right)^2 + b_p \cdot \frac{2}{T}\left(\frac{1 - z^{-1}}{1 + z^{-1}}\right) + \omega_p^2}$$

$$= \frac{2b_pT\,(1 - z^{-1})(1 + z^{-1})}{4(1 - z^{-1})^2 + 2b_pT(1 - z^{-1})(1 + z^{-1}) + \omega_p^2 T^2 (1 + z^{-1})^2}$$

$$= \frac{2b_pT - 2b_pT z^{-2}}{(4 + 2b_pT + \omega_p^2 T^2) + (2\omega_p^2 T^2 - 8) z^{-1} + (4 - 2b_pT + \omega_p^2 T^2) z^{-2}}$$

$$H(z) = \frac{a_0 - a_1 z^{-2}}{1 + b_1 z^{-1} + b_2 z^{-2}}$$

$$\therefore a_0 = a_1 = \frac{2b_pT}{4 + 2b_pT + \omega_p^2 T^2}; \quad b_1 = \frac{2\omega_p^2 T^2 - 8}{4 + 2b_pT + \omega_p^2 T^2}; \quad b_2 = \frac{4 - 2b_pT + \omega_p^2 T^2}{4 + 2b_pT + \omega_p^2 T^2}$$

(b) A digital filter with a centre frequency of 1000 Hz and a bandwidth of 150 Hz is required. Assuming a sampling frequency of 10kHz, compute the digital filter coefficients a_0, a_1, b_1 & b_2 and show that $a_0 = a_1 = 0.04409115$; $b_1 = -1.551$ and $b_2 = 0.918176$.

$\omega_p = 2\pi \times 1000 = 2000\pi$ rad/s

$b_p = 2\pi \times 150 = 300\pi$ rad/s

In order to compensate for non-linear nature of the bilinear transform, it is necessary to prewarp these frequencies:

$$\omega_p' = \frac{2}{T}\tan\left(\omega_p \frac{T}{2}\right) = \frac{2}{10^{-4}}\tan\left(2000\pi \times \frac{10^{-4}}{2}\right) = 6498.4 \text{ rad/s}$$

$$b_p' = \frac{2}{10^{-4}}\tan\left(300\pi \times \frac{10^{-4}}{2}\right) = 943.2 \text{ rad/s}$$

$$\therefore 4 + 2b_pT + \omega_p^2 T^2 = 4 + (2 \times 943.2 \times 10^{-4}) + (6498.4 \times 10^{-4})^2 = 4.610926$$

$$a_0 = a_1 = \frac{2 \times 943.2 \times 10^{-4}}{4.610926} = 0.04409115$$

$$b_1 = \frac{2 \times (6498.4 \times 10^{-4})^2 - 8}{4.610926} = -1.551$$

$$b_2 = \frac{4.233646}{4.610926} = 0.918176$$

(c) Comment on the stability of the digital filter H(z) (see equation 2 above) which you have obtained .

$-2 \leq b_1 = -1.551 \leq 2$

$-1 \leq b_2 = 0.918176 \leq 1$

Stable filter (see stability triangle)

Topic 5: Tutorial questions (SOLO - Level 3)

SOLO – Level 3

2. Using the second order normalised Butterworth lowpass filter G(s), design an IIR digital filter G(z) to approximate the analogue transfer function G(s). The digital filter is to be designed to operate at a sampling rate of 8 kHz with a 3 dB cut-off frequency at 2 kHz. Using the bilinear transformation method, obtain the transfer function G(z) of the digital filter and show that G(z):

$$G(z) = \frac{(z+1)^2}{(2+\sqrt{2})\left(z^2+\frac{2-\sqrt{2}}{2+\sqrt{2}}\right)}$$

$H(s) = \dfrac{1}{s^2+\sqrt{2}s+1}$

$f_c = 2\,kHz$, $f_s = 8\,kHz$

$\theta_c = 2\pi\left(\frac{2000}{8000}\right) = \pi/2$

Prewarp the frequency f_c (ie θ_c)

$\omega_c' = \frac{2}{T}\tan\left(\frac{\theta_c}{2}\right) = \frac{2}{T}\tan\left(\frac{\pi/2}{2}\right) = \frac{2}{T}$ rad/sec

∴ Denormalised transfer function

$$\therefore H(s)\Big|_{s \to \frac{s}{\omega_c'}} = \frac{1}{\left(\frac{s}{\omega_c'}\right)^2+\sqrt{2}\left(\frac{s}{\omega_c'}\right)+1}$$

$$H(s)\Big|_{\text{denormalised}} = \frac{(\omega_c')^2}{s^2+\sqrt{2}\,\omega_c' s+(\omega_c')^2}$$

$\omega_c' = \frac{2}{T}$

$$\therefore H(s) = \frac{\left(\frac{2}{T}\right)^2}{s^2+\frac{2}{T}\sqrt{2}s+\left(\frac{2}{T}\right)^2}$$

$$\therefore H(z) = \frac{\left(\frac{2}{T}\right)^2}{\left(\frac{2}{T}\right)^2\left(\frac{z-1}{z+1}\right)^2+\frac{2}{T}\sqrt{2}\left(\frac{2}{T}\frac{z-1}{z+1}\right)+\left(\frac{2}{T}\right)^2}$$

$$H(z) = \frac{1}{\left(\frac{z-1}{z+1}\right)^2+\sqrt{2}\left(\frac{z-1}{z+1}\right)+1}$$

$$H(z) = \frac{(z+1)^2}{z^2(2+\sqrt{2})+2-\sqrt{2}}$$

$$\therefore H(z) = \frac{(z+1)^2}{(2+\sqrt{2})\left(z^2+\frac{2-\sqrt{2}}{2+\sqrt{2}}\right)}$$

Topic 5: Tutorial questions (SOLO - Level 3)

SOLO – Level 3

3. (a) Determine the impulse response $h_d[n]$ of the bandpass filter whose frequency response is given by

$$H_d(\theta) = \begin{cases} e^{-j3\theta} & \frac{\pi}{4} \le |\theta| \le \frac{3\pi}{4} \\ 0 & \text{otherwise} \end{cases}$$

(b) To obtain a finite impulse response from $h_d[n]$ a Bartlett window of length N = 7 is used. Compute the coefficients of the FIR filter with this impulse response.

Note: The Bartlett Window function is given by

$$w_B[n] = \begin{cases} \frac{2n}{N-1} & 0 \le n < \frac{(N-1)}{2} \\ 2 - \frac{2n}{N-1} & \frac{N-1}{2} \le n \le N-1 \\ 0 & \text{otherwise} \end{cases}$$

$$h_d[n] = \frac{\sin(n-3)\frac{3\pi}{4}}{\pi(n-3)} - \frac{\sin(n-3)\frac{\pi}{4}}{\pi(n-3)} \qquad n \neq 3$$

$$= \frac{1}{2} \quad \text{when } n = 3$$

(b) $N = 7$

$W_B[0] = 0$, $W_B[1] = \frac{1}{3}$, $W_B[2] = \frac{2}{3}$, $W_B[3] = 1$

$W_B[4] = \frac{2}{3}$, $W_B[5] = \frac{1}{3}$, $W_B[6] = 0$ } symmetrical

$\therefore h[n] = h_d[n] \cdot w[n]$

$h[0] = h_d[0] \cdot W_B[0] = 0$

$h[1] = h_d[1] \cdot W_B[1] = -\frac{1}{3\pi}$

$h[2] = h_d[2] \cdot W_B[2] = 0$

$h[3] = h_d[3] \cdot W_B[3] = \frac{1}{2}$ ($h_d[3] = \frac{1}{2}$, $W_B[3] = 1$)

$h[4] = h[2] = 0$

$h[5] = h[1] = -\frac{1}{3\pi}$

$h[6] = h[0] = 0$

$$h_d[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} H_d(\theta) e^{jn\theta} d\theta$$

$$= \frac{1}{2\pi}\int_{-3\pi/4}^{-\pi/4} e^{-j3\theta} \cdot e^{jn\theta} d\theta + \frac{1}{2\pi}\int_{\pi/4}^{3\pi/4} e^{-j3\theta} \cdot e^{jn\theta} d\theta$$

$$= \frac{1}{2\pi}\left[\frac{e^{j(n-3)\theta}}{j(n-3)}\right]_{-3\pi/4}^{-\pi/4} + \frac{1}{2\pi}\left[\frac{e^{j(n-3)\theta}}{j(n-3)}\right]_{\pi/4}^{3\pi/4}$$

$$= \frac{1}{2\pi j(n-3)}\left[e^{-j(n-3)\frac{\pi}{4}} - e^{-j(n-3)\frac{3\pi}{4}} + e^{j(n-3)\frac{3\pi}{4}} - e^{j(n-3)\frac{\pi}{4}}\right]$$

Topic 5: Tutorial questions (SOLO - Level 3)

SOLO – Level 3

4. Using impulse invariant method design a digital filter H(z) to approximate the following normalized analogue transfer function:

$$H(s) = 1/(s^2 + \sqrt{2}\,s + 1)$$

Assuming that the 3dB cut-off frequency of the digital filter is 150Hz and the sampling frequency is 1.28 kHz, Show that H(z) is given by:

$$H(z) = \frac{393.9264z^{-1}}{1 - 1.0308z^{-1} + 0.3530z^{-2}}$$

Before applying the impulse invariant method, we need to denormalise the transfer function.

$$H_1(s) = H(s)\Big|_{s \to \frac{s}{\omega_c}} = \frac{1}{\left(\frac{s}{\omega_c}\right)^2 + \sqrt{2}\left(\frac{s}{\omega_c}\right) + 1}$$

$$H_1(s) = \frac{\omega_c^2}{s^2 + \sqrt{2}\,\omega_c s + \omega_c^2} \quad \text{where } \omega_c = 2\pi \times 150 = 942.4778$$

$$H_1(s) = \frac{\omega_c^2}{s^2 + \sqrt{2}\omega_c s + \omega_c^2} = \frac{A}{s - p_1} + \frac{B}{s - p_2}$$

$$p_1, p_2 = \frac{-\sqrt{2}\omega_c \pm \sqrt{2\omega_c^2 + 4\omega_c^2}}{2}$$

$$= -\sqrt{2}\omega_c\left(\frac{1 \pm j}{2}\right)$$

$$\therefore p_1 = -\frac{\sqrt{2}\omega_c}{2}(1+j) = -666.4324(1+j)$$

$$p_2 = -\frac{\sqrt{2}\omega_c}{2}(1-j) = -666.4324(1-j)$$

Find A & B

$$\Rightarrow A = -\frac{\omega_c}{\sqrt{2}}j$$

$$B = \frac{\omega_c}{\sqrt{2}}j$$

Use impulse invariant transformation

$$H_1(z) = \frac{A}{1 - e^{p_1 T} z^{-1}} + \frac{B}{1 - e^{p_2 T} z^{-1}} \qquad T = \frac{1}{1.28\,kHz}$$

$$= \frac{A(1 - e^{p_2 T} z^{-1}) + B(1 - e^{p_1 T} z^{-1})}{(1 - e^{p_1 T} z^{-1})(1 - e^{p_2 T} z^{-1})}$$

$$= \frac{(A+B) - (A e^{p_2 T} + B e^{p_1 T}) z^{-1}}{1 - (e^{p_1 T} + e^{p_2 T}) z^{-1} + e^{p_1 T} z^{-1}}$$

By substituting for A, B, T, p_1 & p_2

$$H_1(z) = \frac{393.9264\,z^{-1}}{1 - 1.0308\,z^{-1} + 0.3530\,z^{-2}}$$

Topic 5: Tutorial questions (SOLO-Level 4)

SOLO – Level 4

1. Convert an analogue filter $H(s) = \frac{s+0.1}{(s+0.1)^2+16}$ into a digital IIR filter by means of the bilinear transformation. The digital filter is to have a resonant frequency $\theta_0 = \frac{\pi}{2}$.

The analogue filter has a resonant frequency $\omega_0 = 4$ rad/sec

We have $\omega_0 = \frac{2}{T}\tan\frac{\theta_0}{2} \Rightarrow 4 = \frac{2}{T}\tan\left(\frac{\pi/2}{2}\right) \Rightarrow T = \frac{1}{2}$ in order to have $\theta_0 = \pi/2$ rad/sec

Thus the desired mapping is

$$s = \frac{2}{T}\left(\frac{1-z^{-1}}{1+z^{-1}}\right) = 4\left(\frac{1-z^{-1}}{1+z^{-1}}\right)$$

$$H(z) = H(s)\Big|_{s=4\left(\frac{1-z^{-1}}{1+z^{-1}}\right)} = \frac{4\left(\frac{1-z^{-1}}{1+z^{-1}}\right)+0.1}{\left[4\left(\frac{1-z^{-1}}{1+z^{-1}}\right)\right]^2 + 0.2\times 4\left(\frac{1-z^{-1}}{1+z^{-1}}\right) + 16.00}$$

$$H(z) = \frac{0.128 + 0.006z^{-1} - 0.122z^{-2}}{1 + 0.0006z^{-1} + 0.975z^{-2}} = \frac{0.128 + 0.006z^{-1} - 0.122z^{-2}}{1 + 0.975z^{-2}}$$

This filter has Poles at:

$p_1 = 0.987\, e^{j(\pi/2)}$ ← resonant frequency

$p_2 = 0.987\, e^{-j(\pi/2)}$ ← resonant frequency

Topic 5: Tutorial questions (SOLO-Level 4)

SOLO – Level 4

2. Consider the following analogue system with a transfer function $H(s) = \frac{\alpha}{s+\alpha} \Rightarrow h(t) = e^{-\alpha t}$; where α (= 10^4 rad/sec) is the analogue cut-off frequency and the sampling period T is 100 μs.

   (a) Using bilinear transformation, show that the transfer function H(z) is: $H(z) = \frac{1-b}{2}\left[1 + \frac{(1+b)z^{-1}}{1-bz^{-1}}\right]$; where b= $\frac{2-\alpha T}{2+\alpha T}$

   (b) (i) What is the dc gain of H(z)?

   (ii) At what frequency is the H(θ) equal to zero? (θ - digital frequency).

   (iii) Calculate the impulse response h[n].

   (iv) Assuming that the impulse response decays to 1/e of its initial value at n = N samples, show that: N =(ln{b/(b+1)}-1)/(ln(b))

$H(s) = \frac{\alpha}{s+\alpha}$; $H(z) = H(s)\Big|_{s=\frac{2}{T}\frac{1-z^{-1}}{1+z^{-1}}}$

$\therefore H(z) = \frac{\alpha}{\frac{2}{T}\frac{1-z^{-1}}{1+z^{-1}}+\alpha} = \frac{\alpha T(1+z^{-1})}{2(1-z^{-1})+\alpha T(1+z^{-1})}$

$H(z) = \frac{\alpha T(1+z^{-1})}{(2+\alpha T)+(\alpha T-2)z^{-1}} = \frac{\alpha T}{2+\alpha T}\left[\frac{1+z^{-1}}{1-\left(\frac{2-\alpha T}{2+\alpha T}\right)z^{-1}}\right]$

note : Let $b = \frac{2-\alpha T}{2+\alpha T} \Rightarrow 1-b = \frac{2+\alpha T-2+\alpha T}{2+\alpha T}$

$1-b = \frac{2\alpha T}{2+\alpha T}$

$\therefore \frac{1-b}{2} = \frac{\alpha T}{2+\alpha T}$

$H(z) = \frac{1-b}{2}\left[\frac{1+z^{-1}}{1-bz^{-1}}\right] = \frac{1-b}{2}\left[1 + \frac{(1+b)z^{-1}}{1-bz^{-1}}\right]$

$b = \frac{2-10^4\times 100\times 10^{-6}}{2+10^4\times 100\times 10^{-6}} = \frac{1}{3}) \therefore \frac{1-b}{2} = \frac{1}{3}$

$1+b = 4/3$

$H(z) = \frac{1}{3}\left[1 + \frac{\frac{4}{3}z^{-1}}{1-\frac{1}{3}z^{-1}}\right]$

$H(\theta) = \frac{1}{3}\left[1 + \frac{\frac{4}{3}e^{-j\theta}}{1-\frac{1}{3}e^{-j\theta}}\right]$

(i) $H(0) = 1 + \left(\frac{\frac{4}{3}}{1-\frac{1}{3}}\right) = 1$; (ii) $H(\theta)\Big|_{\theta=\pi} = \frac{1}{3}\left[1 + \frac{4e^{-j\pi}}{3-e^{-j\pi}}\right] = 0$

DC gain

$\therefore \theta = \pi$

(iii) Inverse z transform

$h[n] = \frac{1-b}{2}\left[\delta[n] + (1+b)\,b^{n-1}u[n-1]\right]$

(iv) $h[n]\Big|_{n=N} = \frac{1-b}{2}\left[\underbrace{\delta[N]}_{0} + (1+b)\,b^{N-1}\underbrace{u[N-1]}_{=1}\right]$

$h[n] = \frac{1-b}{2}(1+b)\,b^{N-1}$

We have now $\frac{1}{e}h[0] = \frac{1-b}{2}(1+b)\,b^{N-1}$

$\frac{1}{e}\left(\frac{1-b}{2}\right) = \left(\frac{1-b}{2}\right)\cdot(1+b)\,b^{N-1} \Rightarrow \frac{1}{e} = \frac{1+b}{b}\,b^{N}$

$\therefore \ln\left(\frac{1}{e}\right) = \ln\left(\frac{1+b}{b}\right) + N\ln(b)$

$\therefore N\ln(b) = -\ln\left(\frac{1+b}{b}\right) + \ln\left(\frac{1}{e}\right) = \ln\left(\frac{b}{1+b}\right) - 1$

$\therefore N = \frac{\ln\left(\frac{b}{1+b}\right)-1}{\ln(b)}$

Topic 5: Tutorial questions (SOLO-Level 4)

SOLO – Level 4

3. A second-order resonant system is given by $H(s) = \frac{1}{s^2+b_p s+\omega_p^2}$ and $b_p = \frac{\omega_p}{Q_p}$ {ω_p is the resonant frequency, Q_p is the Q - factor associated with the pole , b_p is the bandwidth}. By applying the impulse invariant transformation to the above equation, a digital filter with the transfer function $H(z)$ can be obtained: $H(z) = \frac{k_o z^{-1}}{1+b_1 z^{-1}+b_2 z^{-2}}$. Show that the filter coefficients b_1 and b_2 are given by,

$$b_1 = -2e^{-p_2 T}\cos(q_2 T);\ b_2 = e^{-2p_2 T} \quad \text{where} \quad p_2 = \frac{\omega_p}{2Q_p};\ q_2 = p_2\sqrt{4Q_p^2 - 1}$$

Also show that $k_o = -4q_2 e^{-p_2 T}\sin(q_2 T)$ { T is the sampling period}

$$H(s) = \frac{1}{s^2 + \frac{\omega_p}{Q_p}s + \omega_p^2} = \frac{1}{s^2 + b_p s + \omega_p^2}$$

$$s_{1,2} = \frac{-b_p \pm \sqrt{b_p^2 - 4\omega_p^2}}{2} = -\frac{b_p}{2} \pm j\sqrt{4\omega_p^2 - b_p^2}$$

$$= \frac{-\omega_p}{2Q_p} \pm j\frac{\omega_p}{2Q_p}\sqrt{4Q_p^2 - 1}$$

$$s_{1,2} = -p_2 \pm j q_2 \ ; \quad p_2 = \frac{-\omega_p}{2Q_p}; \quad q_2 = p_2\sqrt{4Q_p^2 - 1}$$

$$H(s) = \frac{1}{(s-s_1)(s-s_2)} = \frac{A}{s-s_1} + \frac{B}{s-s_2} = \frac{\frac{1}{s_1-s_2}}{s-s_1} + \frac{\frac{-1}{s_1-s_2}}{s-s_2}$$

$$H(s) = \frac{1}{s_1-s_2}\left[\frac{1}{s-s_1} - \frac{1}{s-s_2}\right]$$

↓ impulse invariant transform

$$H(z) = \frac{1}{s_1-s_2}\left[\frac{1}{1-e^{-p_2 T + j q_2 T}z^{-1}} - \frac{1}{1-e^{-p_2 T - j q_2 T}z^{-1}}\right]$$

$$H(z) = \frac{1}{s_1-s_2}\ \frac{z^{-1}\left(e^{jq_2T} - e^{-jq_2T}\right)e^{-p_2T}}{1 - 2e^{-p_2T}z^{-1}\left(\frac{e^{jq_2T}+e^{-jq_2T}}{2}\right) + e^{-2p_2T}z^{-2}}$$

($\cos(q_2T)$; $\sin(q_2T)$)

$$H(z) = 2jq_2\ \frac{z^{-1}(\ )e^{-p_2T}}{1 + b_1 z^{-1} + b_2 z^{-2}}$$

$$= \frac{-4q_2 e^{-p_2T} z^{-1}\left(\frac{e^{+jq_2T} - e^{-jq_2T}}{2j}\right)}{1 + b_1 z^{-1} + b_2 z^{-2}}$$

$$= \frac{K_0 z^{-1}}{1 + b_1 z^{-1} + b_2 z^{-2}}$$

$$K_0 = -4q_2 e^{-p_2T}\sin(q_2T)$$

$$b_1 = -2e^{-p_2T}\cos(q_2T)$$

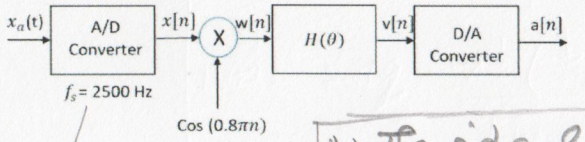
$$b_2 = e^{-2p_2T}$$

Topic 5: Tutorial questions (SOLO-Level 5)

SOLO – Level 5

1. An analogue signal of the form $x_a(t) = a(t) \cos 2000\pi t$ is bandlimited to the range $900 \le f \le 1100\ Hz$. It is used as an input to the system shown below:
   (a) Determine and sketch the spectra for the signal x[n] and w[n]. $W_H[n] = 0.54 - 0.46 \cos \frac{2\pi(n-3)}{6}$ [given]
   (b) Use a Hamming window of length M= 7 to design a lowpass linear phase FIR filter $H(\theta)$ that passes a[n]

$f_c = 1000\ Hz = \frac{1000}{2500} = 0.4 \} \theta_c = 0.8\pi$

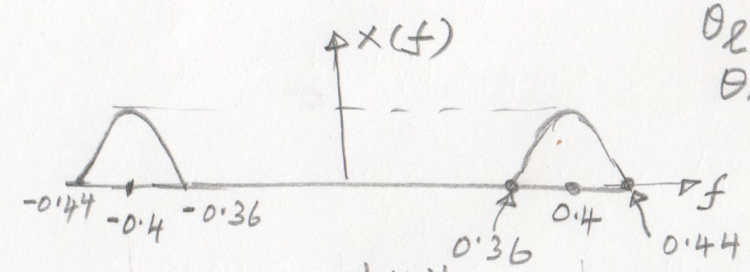


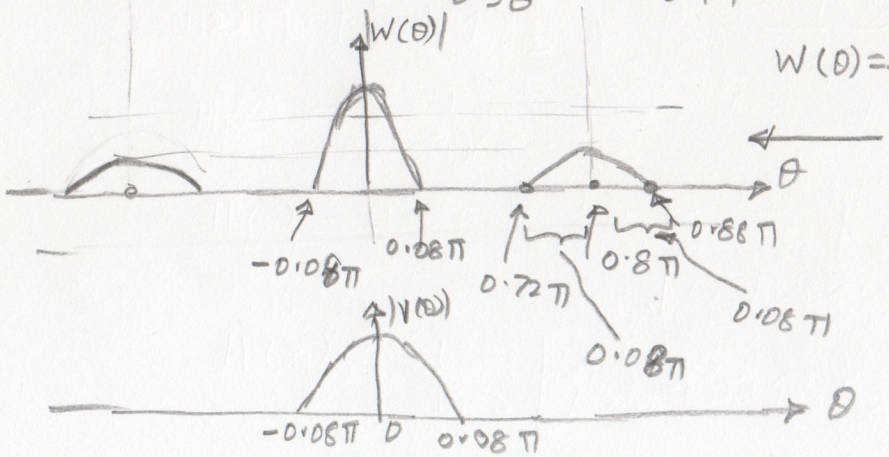
$x_a(t) = a(t) \cos 2000\pi t$

$f_l = \frac{900}{2500} = 0.36$; $f_h = \frac{1100}{2500} = 0.44$

$\theta_l = 0.72\pi$

$\theta_h = 0.88\pi$





$W(\theta) = \frac{1}{2}\left[X(\theta-\theta_0) + X(\theta-\theta_0)\right]$

see Page 21 of lecture notes

(b) linear phase filter:

$M = 7$

Symmetry around $n = 3$

(b) The ideal lowpass filter has a passband of $-0.08\pi \le \theta \le 0.08\pi$,

Hence

$$H_d(\omega) = \begin{cases} 1 \cdot e^{-j(3\theta)} & -0.08\pi \le \theta \le 0.08\pi \\ 0 & \text{otherwise} \end{cases}$$

$$h_d[n] = \frac{1}{2\pi} \int_{-0.08\pi}^{0.08\pi} e^{-j3\theta} \cdot e^{jn\theta}\, d\theta$$

$$= \frac{1}{2\pi} \int_{-0.08\pi}^{0.08\pi} e^{j\theta(n-3)}\, d\theta$$

$$\boxed{h_d[n] = 2\,\frac{\sin 0.08\pi(n-3)}{(n-3)}}$$

$\therefore h[n] = h_d[n] \cdot W_H[n]$

$\left[W_H[n] = 0.54 - 0.46\left\{\cos\frac{2\pi(n-3)}{6}\right\}\right]$ given

$h[n]$ is the impulse response of the lowpass filter $H(\theta)$.

Topic 5: Tutorial questions (SOLO-Level 5)

SOLO – Level 5

2. A continuous time system is called an integrator if the response of the system $y_a(t)$ to an input $x_a(t)$ is :

$$y_a(t) = \int_{-\infty}^{t} x_a(\tau)d\tau$$

The system function for an integrator is $H_a(s) = \frac{1}{s}$

(a) Design a discrete –time integrator using bilinear transformation and find the difference equation relating the input x[n] to the output y[n] of the discrete-time system.

(b) Find the frequency response of the discrete-time integrator found in part (a) , and determine whether or not this system is a good approximation to the continuous-time system.

Bilinear Transform :

$s = \frac{2}{T}\frac{1-z^{-1}}{1+z^{-1}}$

with a system function $H_a(s) = \frac{1}{s}$,

the bilinear transform produces a discrete-time filter with a system function

$$H(z) = \frac{T}{2}\frac{1+z^{-1}}{1-z^{-1}}$$

$$\frac{Y(z)}{X(z)} = \frac{T}{2}\frac{1+z^{-1}}{1-z^{-1}}$$

$$Y(z)[1-z^{-1}] = \frac{T}{2}(1+z^{-1})X(z)$$

$$\therefore y[n] = \frac{T}{2}\{x[n]+x[n-1]\} + y[n-1]$$

(b) Because the frequency response of the continuous-time system $H_a(j\omega) = \frac{1}{j\omega}$ is related to the discrete-time filter through the mapping $\omega = \frac{2}{T}\tan(\theta/2)$

the frequency response of the discrete-time system is

$$H(\theta) = H_a(j\omega)\Big|_{\omega=\frac{2}{T}\tan\theta/2} = \frac{1}{j\cdot\frac{2}{T}\tan\theta/2}$$

$$H(\theta) = \frac{T}{2j}\cot\theta/2 \quad \& \; H_a(j\omega) = \frac{1}{j\omega}$$

$$H(\theta) = \frac{T}{2j}\frac{\cos\theta/2}{\sin\theta/2}$$

for small θ, $H(\theta) = \frac{T}{2j}\cdot\frac{1}{\theta/2}$ $(\theta \approx 0)$ small θ

$$= \frac{T}{2j}\cdot\frac{1}{\omega T/2}$$

$$= \frac{1}{\omega j}$$

$$\therefore H(\theta) \approx H_a(j\omega)$$